Whole-body cell references for mapping genetic risk

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01 Public single-cell data as a reference for genetic risk

scBaseCount provides over 502 million cells across organisms. [1] We use human primary samples to build tissue–cell expression references and test associations with genome-wide association study (GWAS) results.

02 Curation and cell annotation

AI-assisted metadata curation identifies primary samples. Pan-human Azimuth assigns cell labels from gene expression. [1,2]

The human fine reference retains labels with annotation scores ≥0.90. A score threshold does not establish annotation accuracy.

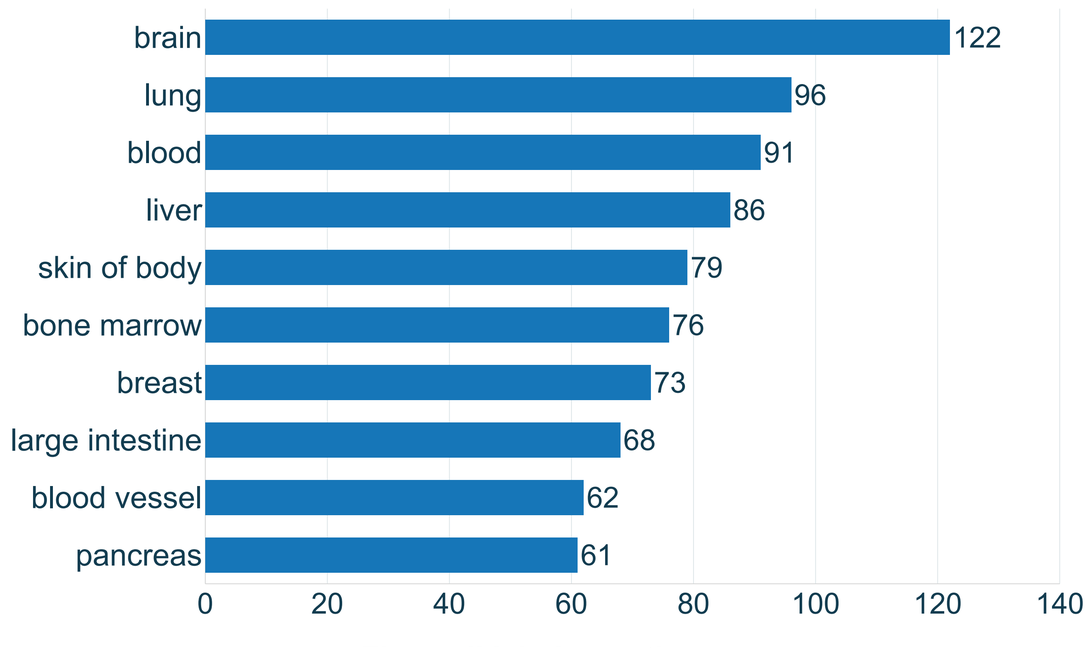
03 The reference used here

50,350,426 cells across 2,408 tissue–cell populations in the 28 October snapshot.

Gene specificity measures expression relative to the reference background. MAGMA tests its association with genetic risk. [3,4]

04 Coverage varies by tissue

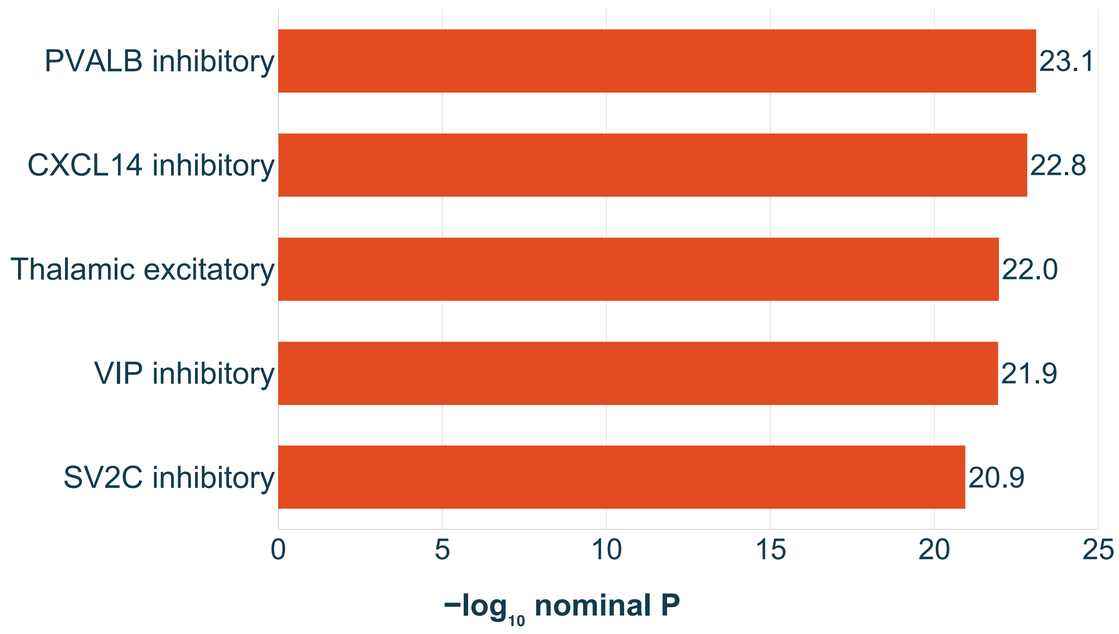
55 tissue labels; 2,408 tested populations
Ten tissues with the most fine cell labels



The remaining 45 tissue labels contain 1594 populations. Counts describe reference coverage, not cell abundance or annotation accuracy.

05 Schizophrenia example

Brain PVALB inhibitory neurons rank first among 2,408 tissue–cell populations. [5]



Top five populations, all labelled brain.
PVALB: $P = 8.04 \times 10^{-24}$; adjusted $P = 1.16 \times 10^{-19}$.
Significance: $-\log_{10}(P) > 5.46$ ($\alpha = 0.05$).

Continuous adjusted MAGMA model; native population background. Bonferroni correction across 2,408 populations × 6 studies. [3–5]

06 A large reference with testable disease associations

This reference links public single-cell data to genetic association signals. In schizophrenia, the strongest signal maps to brain PVALB inhibitory neurons.

Uneven tissue coverage and uncertain annotations limit interpretation. Next: test finer populations in independent data.

[1] scBaseCount, Arc Institute. January 2026 release.
[2] Pan-human Azimuth. doi:10.1101/2026.07.16.738997
[3] Skene et al. Nature Genetics (2018). doi:10.1038/s41588-018-0129-5
[4] de Leeuw et al. PLoS Comput Biol (2015). doi:10.1371/journal.pcbi.1004219
[5] Trubetskoy et al. Nature (2022). doi:10.1038/s41586-022-04434-5

Methods & software
neurogenomics.github.io/MAGMA_Celltyping/